

SOLO® ADSS Medium-Span, Loose Tube, Gel-Filled Cable 12 F, 50 µm multimode (OM4)

CORNING

Part Number: 012TN4-T4M90A20

Corning SOLO® ADSS medium-span cables are all-dielectric, self-supporting (ADSS) cables designed for easy and economical one-step installation in campus backbones with self-supporting installations where metallic messengers cannot be used. The loose tube design provides stable performance over a wide temperature range and is compatible with any telecommunications-grade optical fiber. The economical single-jacket design can span distances of 800 ft in NESC light conditions, 650 ft in NESC medium conditions and 450 ft in NESC heavy conditions. This cable incorporates innovative waterblocking materials, eliminating the need for traditional flooding compound and providing efficient and craft-friendly cable preparation. While the concentric, self-supporting cable design allows easy, one-step installation using standard hardware and installation methods, the SZ-stranded, loose tube design isolates optical fibers from installation and environmental rigors and facilitates mid-span access. The ADSS optical cables are also available with a proprietary track-resistant polyethylene (TRPE) jacket suitable for installation in electric field potentials up to 25 kV.

Features and Benefits

Loose tube design

Stable performance and compatibility with all common fiber types

Self-supporting

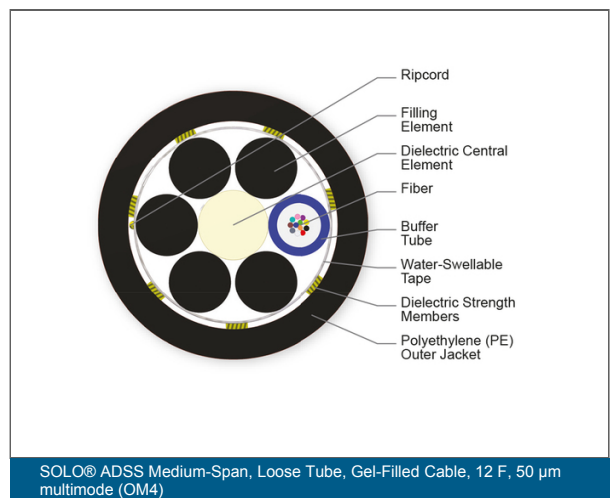
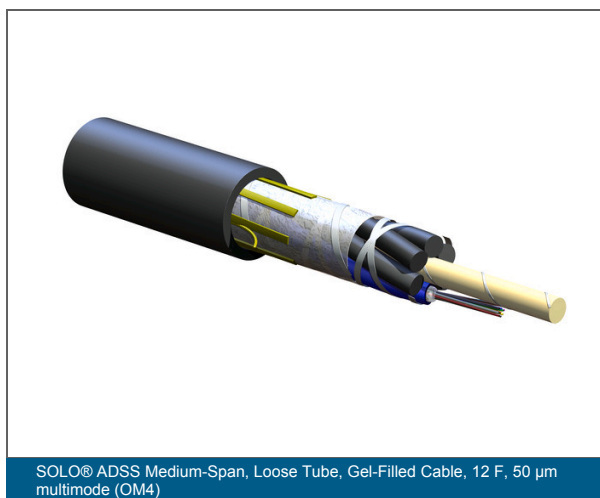
Easy, one-step installation

Track-resistant jacket available

Suitable for installations up to 25 kV electric field potential

Innovative waterblocking cable core

Provides efficient and craft-friendly cable preparation



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Specifications

General Specifications

Cable Type	Loose Tube
Environment	Outdoor
Product Type	Self-Supporting, ADSS
Fiber Category	50 µm MM (OM4)
Cable geometry	Other

Standards

RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Approvals and Listings	USDA Rural Development Programs
Common Installations	Outdoor self-supporting aerial
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions

Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-40 °C to 70 °C (-40 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design

Central Element	Dielectric
Fiber Count	12
Number of Ripcords	1
Outer Jacket Color	Black
Outer Jacket Material	Polyethylene (PE)
Tensile Strength Elements and/or Armoring - Layer 1	Dielectric strength members
Buffer Tube Color	Blue
Buffer Tube Diameter	2.5 mm (0.1 in)

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Cable Design

Number of Active Tubes	1
Number of Filling Elements	5
Number of Tube Positions	6
Tape	Water-swellable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12
Color Code Standards	Telcordia

Mechanical Specifications

Nominal Outer Diameter	11.3 mm (0.44 in)
Min. Bend Diameter Installation	340 mm (13.39 in)
Min. Bend Diameter Operation	226 mm (8.9 in)

Optical Characteristics

Fiber Code	T
Performance Option Code	90
Fiber Category	OM4
Fiber Type	Multimode
Fiber Name	50 µm MM (OM4)
Maximum Attenuation	3.0 dB/km / 1.0 dB/km
Wavelengths	850 nm / 1300 nm
Fiber Compliance	IEC 60793-2-10
Fiber Core Diameter	50 µm
Serial 1 Gigabit Ethernet	1000 m / 600 m
Serial 10 Gigabit Ethernet	550 m / -
Minimum Effective Modal Bandwidth (EMB)	4700 MHz*km / -
Min. Overfilled Launch (OFL) Bandwidth	3500 MHz*km / 500 MHz*km

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Dimensions

Cable Weight	97 kg/km (65.18 lb/1000 ft)
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Installation Characteristics

	Initial Installation		NESC Light		NESC Medium		NESC Heavy	
Span	SAG	Tension	SAG	Tension	SAG	Tension	SAG	Tension
Fiber Count 12								
15 m (49.2 ft)	1 %	181 N (40.7 lbf)	0.4 %	495 N (111.3 lbf)	1.6 %	717 N (161.2 lbf)	2.3 %	1,092 N (245.5 lbf)
30 m (98.4 ft)	1 %	362 N (81.4 lbf)	0.5 %	893 N (200.8 lbf)	1.9 %	1,256 N (282.4 lbf)	2.7 %	1,862 N (418.6 lbf)
46 m (150.9 ft)	1 %	543 N (122.1 lbf)	0.5 %	1,249 N (280.8 lbf)	2.1 %	1,728 N (388.5 lbf)	2.9 %	2,527 N (568.1 lbf)
61 m (200.1 ft)	1 %	725 N (163 lbf)	0.5 %	1,580 N (355.2 lbf)	2.2 %	2,162 N (486 lbf)	3.2 %	3,133 N (704.3 lbf)
76 m (249.3 ft)	1 %	906 N (203.7 lbf)	0.6 %	1,894 N (425.8 lbf)	2.3 %	2,569 N (577.5 lbf)	3.4 %	3,698 N (831.3 lbf)
91 m (298.5 ft)	1 %	1,087 N (244.4 lbf)	0.6 %	2,195 N (493.5 lbf)	2.4 %	2,956 N (664.5 lbf)	3.5 %	4,233 N (951.6 lbf)
107 m (351 ft)	1 %	1,268 N (285.1 lbf)	0.6 %	2,485 N (558.6 lbf)	2.5 %	3,328 N (748.2 lbf)	3.7 %	4,745 N (1,066.7 lbf)
122 m (400.2 ft)	1 %	1,449 N (325.8 lbf)	0.6 %	2,768 N (622.3 lbf)	2.6 %	3,688 N (829.1 lbf)	3.8 %	5,238 N (1,177.5 lbf)
137 m (449.4 ft)	1 %	1,630 N (366.4 lbf)	0.6 %	3,043 N (684.1 lbf)	2.7 %	4,037 N (907.5 lbf)	3.9 %	5,715 N (1,284.8 lbf)
152 m (498.6 ft)	1 %	1,811 N (407.1 lbf)	0.6 %	3,313 N (744.8 lbf)	2.7 %	4,377 N (984 lbf)	4.1 %	6,179 N (1,389.1 lbf)
168 m (551 ft)	1 %	1,992 N (447.8 lbf)	0.6 %	3,577 N (804.1 lbf)	2.8 %	4,710 N (1,058.8 lbf)	4.2 %	6,631 N (1,490.7 lbf)
183 m (600.2 ft)	1 %	2,174 N (488.7 lbf)	0.7 %	3,837 N (862.6 lbf)	2.8 %	5,036 N (1,132.1 lbf)	4.3 %	7,073 N (1,590.1 lbf)

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Span	SAG	Tension	SAG	Tension	SAG	Tension	SAG	Tension
198 m (649.4 ft)	1 %	2,355 N (529.4 lbf)	0.7 %	4,093 N (920.1 lbf)	2.9 %	5,355 N (1,203.8 lbf)	4.4 %	7,506 N (1,687.4 lbf)
213 m (698.6 ft)	1 %	2,536 N (570.1 lbf)	0.7 %	4,346 N (977 lbf)	2.9 %	5,670 N (1,274.7 lbf)	4.4 %	7,931 N (1,783 lbf)
229 m (751.1 ft)	1 %	2,717 N (610.8 lbf)	0.7 %	4,595 N (1,033 lbf)	3 %	5,980 N (1,344.4 lbf)		
244 m (800.3 ft)	1 %	2,898 N (651.5 lbf)	0.7 %	4,842 N (1,088.5 lbf)	3 %	6,285 N (1,412.9 lbf)		
259 m (849.5 ft)	1 %	3,079 N (692.2 lbf)	0.7 %	5,086 N (1,143.4 lbf)	3.1 %	6,586 N (1,480.6 lbf)		
274 m (898.7 ft)	1 %	3,260 N (732.9 lbf)	0.7 %	5,328 N (1,197.8 lbf)	3.1 %	6,884 N (1,547.6 lbf)		
290 m (951.2 ft)	1 %	3,441 N (773.6 lbf)	0.7 %	5,567 N (1,251.5 lbf)	3.2 %	7,178 N (1,613.7 lbf)		
305 m (1,000.4 ft)	1 %	3,623 N (814.5 lbf)	0.7 %	5,804 N (1,304.8 lbf)	3.2 %	7,469 N (1,679.1 lbf)		
320 m (1,049.6 ft)	1 %	3,804 N (855.2 lbf)	0.7 %	6,040 N (1,357.8 lbf)	3.2 %	7,757 N (1,743.8 lbf)		
335 m (1,098.8 ft)	1 %	3,985 N (895.9 lbf)	0.7 %	6,274 N (1,410.5 lbf)	3.3 %	8,043 N (1,808.1 lbf)		
351 m (1,151.3 ft)	1 %	4,166 N (936.5 lbf)	0.7 %	6,506 N (1,462.6 lbf)				
366 m (1,200.5 ft)	1 %	4,347 N (977.2 lbf)	0.8 %	6,737 N (1,514.5 lbf)				
381 m (1,249.7 ft)	1 %	4,528 N (1,017.9 lbf)	0.8 %	6,966 N (1,566 lbf)				
396 m (1,298.9 ft)	1 %	4,709 N (1,058.6 lbf)	0.8 %	7,194 N (1,617.3 lbf)				

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Span	SAG	Tension	SAG	Tension	SAG	Tension	SAG	Tension
411 m (1,348.1 ft)	1 %	4,891 N (1,099.5 lbf)	0.8 %	7,421 N (1,668.3 lbf)				
427 m (1,400.6 ft)	1 %	5,072 N (1,140.2 lbf)	0.8 %	7,646 N (1,718.9 lbf)				
442 m (1,449.8 ft)	1 %	5,253 N (1,180.9 lbf)	0.8 %	7,871 N (1,769.5 lbf)				
457 m (1,499 ft)	1 %	5,434 N (1,221.6 lbf)	0.8 %	8,094 N (1,819.6 lbf)				



Corning Optical Communications LLC • 4200 Corning Place • Charlotte, NC • 28216 • United States
800-743-2675 • FAX: 828-325-5060 • International: +1-828-901-5000 • www.corning.com/opcomm

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